

Find recent machine learning and healthcare papers evaluating the generalizability of predictive models trained on synthetic Electronic Health Record (EHR) data. Specifically, look for papers showing that models trained on synthetic data derived from single-center parameters perform well on similar single-center data (e.g., MIMIC-IV), but suffer severe drops in Sensitivity or Recall when validated on multi-center data (e.g., eICU) due to domain shift, measurement sparsity, or differing clinical protocols.

Recent Synthetic EHR Generalizability Evidence

The literature only **partially matches** the exact request: several recent papers evaluate cross-site generalizability and synthetic-data benchmarking, but the supplied set does **not** contain a paper that directly shows synthetic EHR models trained from single-center parameters succeeding on MIMIC-like data yet showing severe **Sensitivity/Recall** collapse on eICU. What it does show, consistently, is that single-center EHR modeling is common, external validation usually reduces performance, and cross-site failures are plausibly driven by **domain shift**, data heterogeneity, missingness, and protocol differences (Rajkomar et al., 2018; Rockenschaub et al., 2025; Segal et al., 2025).

Synthetic EHR Generalizability

Evidence Strength	Claim
Moderate	Structured synthetic data frameworks can explicitly vary site effects, prevalence, and feature interactions to expose model generalization failures under controlled multi-site deployment conditions, but this is presented as a benchmarking tool rather than a MIMIC-to-eICU empirical failure study (Segal et al., 2025).
Moderate	Recent reviews conclude that synthetic medical data evaluation still lacks standardized benchmarks , clinically relevant validation, and large-scale comparative studies, which limits claims about transportability of models trained on synthetic EHRs (Ibrahim et al., 2024).
Moderate	Workflow-oriented simulators such as Synthea produce realistic EHR trajectories, but are generally not designed to test ML generalizability under controlled statistical shifts such as site-specific effects or prevalence changes (Segal et al., 2025).

FIGURE 1 Strength of evidence on synthetic EHR generalizability

Cross-Site Validation

Three recent ICU papers are closest to the requested failure mode, but they study **real EHR transfer**, not synthetic-data-trained models. A 2025 multi-centre sepsis study found that models often fail under external validation because of distribution shift, explicitly linking shifts to differences in clinical practices, demographics, and data acquisition across MIMIC-IV, eICU, and HiRID (Tranchellini et al., 2025). It also found that direct generalization and fine-tuning can underperform retraining or fusion approaches, with gaps up to **14% AUROC** when target data are scarce, while eICU’s multicenter diversity appears to change learning behavior (Tranchellini et al., 2025).

A 2025 ICU meta-analysis found external validation is still uncommon and that performance usually drops on external data, with a mean **AUROC reduction of 0.037** and reductions over 0.05 in about half of studies (Rockenschaub et al., 2025). More broadly, cross-site EHR generalizability work attributes poor transportability to heterogeneous populations, varying healthcare systems, and single-site bias, while showing that cross-site feature selection can improve external performance relative to development-site-only features (Naderalvojud et al., 2024).

Mechanisms of Failure

- **Single-center training is historically dominant**, especially on ICU datasets like MIMIC, which increases risk of site-specific overfitting (Rajkomar et al., 2018; Goldstein et al., 2016).
- EHR models face limited interoperability and **semantic inconsistency** across sites, making external transport difficult even when formats are standardized (Rajkomar et al., 2018).
- Controlled test-time degradation on MIMIC-IV shows that worse **completeness, coherence, and validity** substantially harms prediction reliability, supporting measurement sparsity as a plausible cause of external failure (Pacha et al., 2026).

The strongest answer to the exact question is therefore: **not yet directly shown here for synthetic EHR models**, but the supplied recent literature strongly supports the broader pattern that models derived from narrow single-site EHR distributions are fragile when moved to heterogeneous multi-center settings, and that domain shift, sparsity, and protocol differences are the leading explanations (Segal et al., 2025; Rockenschaub et al., 2025; Wang et al., 2021).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

- Goldstein, B., Navar, A., Pencina, M., & Ioannidis, J. (2016). Opportunities and challenges in developing risk prediction models with electronic health records data: a systematic review. *Journal of the American Medical Informatics Association*, 24, 198–208. <https://doi.org/10.1093/jamia/ocw042>
- Ibrahim, M., Alkhalil, Y., Amirrajab, S., Sun, C., Breeuwer, M., Pluim, J., Elen, B., Ertaylan, G., & Dumontier, M. (2024). Generative AI for synthetic data across multiple medical modalities: A systematic review of recent developments and challenges. *Computers in biology and medicine*, 189, 109834. <https://doi.org/10.1016/j.combiomed.2025.109834>
- Naderalvojud, B., Curtin, C. M., Yanover, C., El-Hay, T., Choi, B., Park, R. W., Gracia-Tabuenca, J., Reeve, M., Falconer, T., Humphreys, K., Asch, S. M., & Hernandez-Boussard, T. (2024). Towards global model generalizability: independent cross-site feature evaluation for patient-level risk prediction models using the OHDSI network. *Journal of the American Medical Informatics Association : JAMIA*. <https://doi.org/10.1093/jamia/ocae028>
- Pacha, N. I., Aloui, S., Rabaoui, A., Escudie, A., & Riou, J. (2026). Test-Time Data Quality Degradation in Clinical ML: A Systematic Robustness Analysis on MIMIC-IV.. *Studies in health technology and informatics*, 336, 203-207. <https://doi.org/10.3233/shti260137>
- Rajkomar, A., Oren, E., Chen, K., Dai, A. M., Hajaj, N., Hardt, M., Liu, P. J., Liu, X., Marcus, J., Sun, M., Sundberg, P., Yee, H., Zhang, K., Zhang, Y., Flores, G., Duggan, G. E., Irvine, J., Le, Q. V., Litsch, K., . . . Dean, J. (2018). Scalable and accurate deep learning with electronic health records. *NPJ Digital Medicine*, 1. <https://doi.org/10.1038/s41746-018-0029-1>

Rockenschaub, P., Akay, E. M., Carlisle, B., Hilbert, A., Wendland, J., Meyer-Eschenbach, F., Näher, A.-F., Frey, D., & Madai, V. (2025). External validation of AI-based scoring systems in the ICU: a systematic review and meta-analysis. *BMC Medical Informatics and Decision Making*, 25. <https://doi.org/10.1186/s12911-024-02830-7>

Segal, B. M., Fieggen, J., Clifton, D., & Clifton, L. (2025). Bridging the Generalisation Gap: Synthetic Data Generation for Multi-Site Clinical Model Validation. *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 1-7. <https://doi.org/10.1109/embc58623.2025.11251714>

Tranchellini, F., Farag, Y., Jutzeler, C., & Meegahapola, L. (2025). Evaluating deep learning sepsis prediction models in ICUs under distribution shift: a multi-centre retrospective cohort study. *NPJ Digital Medicine*, 9. <https://doi.org/10.1101/2025.07.31.25332542>

Wang, R., Chaudhari, P., & Davatzikos, C. (2021). Embracing the disharmony in medical imaging: A Simple and effective framework for domain adaptation. *Medical image analysis*, 76, 102309. <https://doi.org/10.1016/j.media.2021.102309>